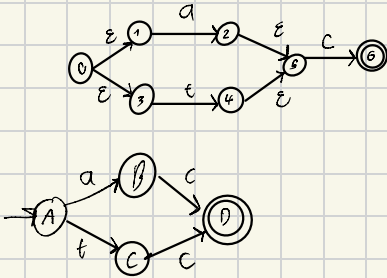


Ejercicio No. 1 (25%) – Convierta las siguientes expresiones regulares en autómatas finitos deterministas (para ello deberá primero convertir las expresiones regulares a AFN y luego convertir a AFD). Muestre todo su procedimiento, i.e., AFN construido con Thompson, tabla de transición, conversión a AFD. Para el inciso g, interprete \ como un escape de carácter, i.e., \ significa que su regex reconoce el carácter \.

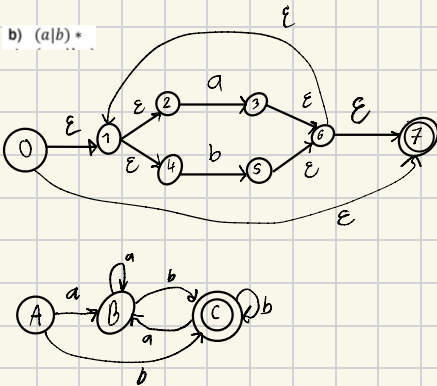
a) $(a|t)c$



	a	t	c	ε
0	✓	✓	✓	{0, 3}
1	2	✓	✓	{1}
2	✓	✓	✓	{2, 5}
3	✓	4	✓	{3}
4	✓	✓	✓	{4, 5}
5	✓	✓	6	{5}
6	✓	✓	✓	{6}

Estado AFN	aε	tε	cε*
A = {0, 3}	B	C	✓
B = {2, 5}	✓	✓	D
C = {4}	✓	✓	D
D = {6}	✓	✓	✓

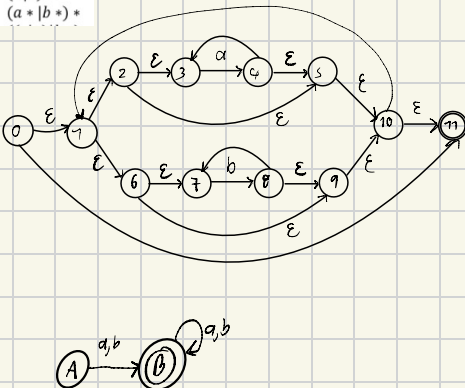
b) $(a|b)^*$



	a	b	ε
0	✓	✓	{0, 3}
1	✓	✓	{1, 3, 4}
2	3	✓	{2}
3	✓	✓	{3, 6}
4	✓	{5}	{4}
5	✓	✓	{3, 6}
6	✓	✓	{6, 7, 3}
7	✓	✓	{7}

Estado AFN	aε	bε*
A = {0, 3, 4, 7}	B	C
B = {1, 2, 3, 4, 6, 3}	B	C
C = {1, 2, 4, 5, 6, 3}	B	C

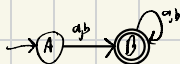
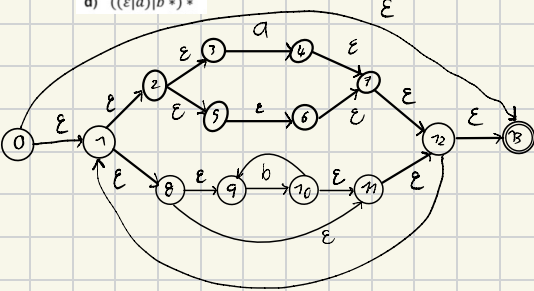
c) $(a^*|b^*)^*$



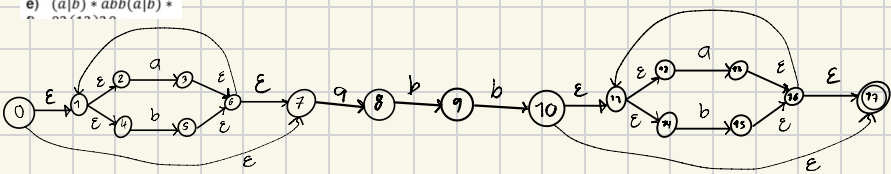
	a	b	ε
0	✓	✓	{1}
1	✓	✓	{1, 3, 7}
2	✓	✓	{2, 3, 5}
3	4	✓	{3}
4	✓	✓	{3, 4, 5}
5	✓	✓	{5, 10}
6	✓	✓	{6, 7, 9}
7	✓	8	{7}
8	✓	✓	{8, 7, 9}
9	✓	✓	{9, 10}
10	✓	✓	{10, 11, 3}
11	✓	✓	{11}

Estado AFN	aε	bε*
A = {0, 1, 3, 4, 5, 6, 7, 9, 10, 11}	B	B
B = {0, 3, 3, 5, 1, 3, 5, 7, 10, 11}	B	B

d) $((\epsilon|a)|b)^*$



e) $(a|b)^*abb(a|b)^*$



Estado AFN

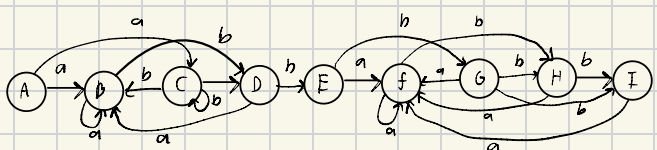
	$a\epsilon^*$	$b\epsilon^*$
$A = \{0, 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$	B	B
$B = \{0, 1, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$	B	B

	a	b	ϵ
0	/	/	$\{0, 1, 13\}$
1	/	/	$\{1, 2, 5\}$
2	/	/	$\{2, 3, 6\}$
3	/	/	$\{3\}$
4	/	/	$\{4, 7\}$
5	/	/	$\{5, 6\}$
6	/	/	$\{6, 9\}$
7	/	/	$\{7, 12\}$
8	/	/	$\{8, 9, 11\}$
9	/	10	$\{9\}$
10	/	/	$\{9, 10, 11\}$
11	/	/	$\{11, 12\}$
12	/	/	$\{12, 13, 11\}$
13	/	/	$\{13\}$

Tabla AFD

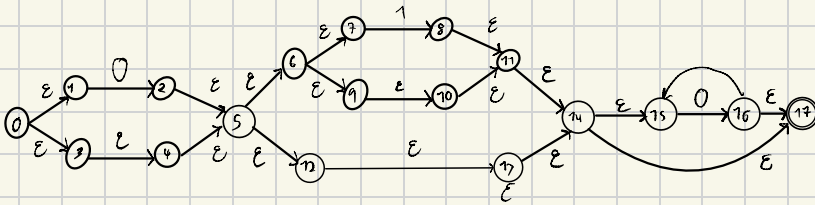
	a	b	ϵ
0	/	/	$\{0, 13\}$
1	-	-	$\{1, 3, 4\}$
2	3	/	$\{2\}$
3	/	/	$\{3, 6\}$
4	/	5	$\{4\}$
5	/	/	$\{5, 6\}$
6	/	/	$\{6, 7\}$
7	8	/	$\{7\}$
8	/	9	$\{8\}$
9	/	10	$\{9\}$
10	/	/	$\{10, 11\}$
11	/	/	$\{11, 12, 14\}$
12	13	/	$\{12\}$

Estados	conjuntos	$a\epsilon^*$	$b\epsilon^*$
A	$\{0, 1, 2, 4, 13\}$	B	C
B	$\{1, 2, 3, 4, 6, 7, 8\}$	B	D
C	$\{1, 2, 4, 5, 6, 13\}$	B	C
D	$\{1, 2, 4, 5, 6, 7, 9\}$	B	E
E	$\{1, 2, 4, 5, 6, 7, 10, 11, 12, 14, 15\}$	F	G
F	$\{1, 2, 3, 4, 6, 7, 8, 11, 12, 13, 14, 15, 16, 17\}$	F	H
G	$\{1, 2, 4, 5, 6, 7, 11, 12, 14, 15, 16, 17\}$	F	H
H	$\{1, 2, 4, 5, 6, 7, 9, 11, 12, 14, 15, 16, 17\}$	F	I
I	$\{1, 2, 4, 5, 6, 7, 10, 11, 12, 14, 15, 16, 17\}$	F	G



7) $0?(1?)?0*$

$(\epsilon|0)(\epsilon|(\epsilon|1))0^*$

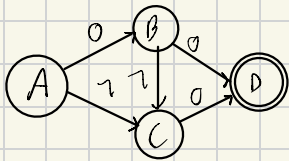


	0	1	ϵ
0	/	/	$\{0, 1, 3\}$
1	2	/	$\{1\}$
2	/	/	$\{2, 5\}$
3	/	/	$\{3, 4\}$
4	/	/	$\{4, 5\}$
5	/	/	$\{5, 6, 12\}$
6	/	/	$\{6, 7, 9\}$
7	/	8	$\{7\}$
8	/	/	$\{8, 13\}$
9	/	/	$\{9, 10\}$
10	/	/	$\{10, 11\}$
11	/	/	$\{11, 14\}$
12	/	/	$\{12, 17\}$

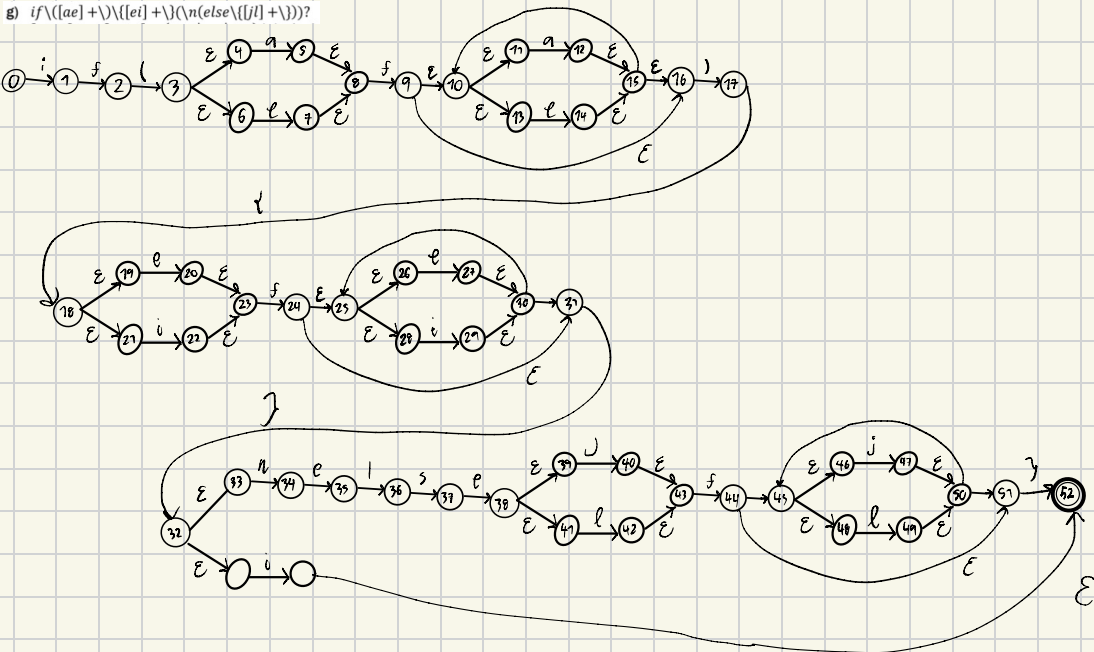
	0	1	ϵ
13	/	/	$\{13, 14\}$
14	/	/	$\{14, 15\}$
15	16	/	$\{15\}$
16	/	/	$\{16, 18\}$
17	/	/	$\{17\}$

Tabla AFD

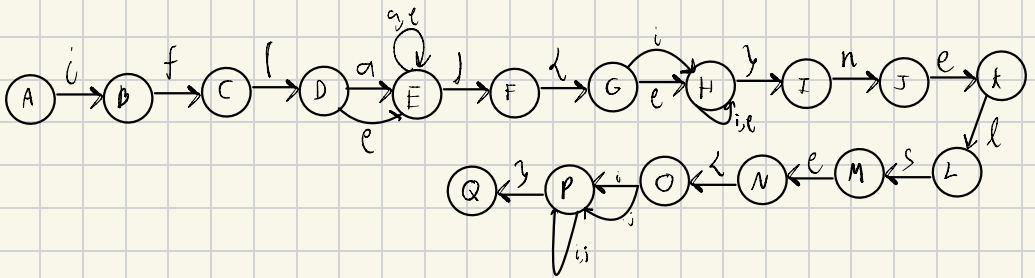
Estados	Conjuntos	$0\epsilon^*$	$1\epsilon^*$
A	$\{0, 1, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14\}$	B	C
B	$\{2, 5, 6, 7, 9, 10, 11, 12, 13, 14\}$	D	C
C	$\{8, 13, 14\}$	D	
D	$\{14\}$		



g) *if\([ae] + \backslash\{[ei] + \backslash\}(\backslash n(\text{else}\backslash\{[jl] + \backslash\}))\)?*

[illegible]

Conjuntos	ae*	ee*	ie*	je*	le*
{1, 3, 5, 20, 28, 30, 39, 40, 41, 42, 43, 44}	A	C	D	E	F
{2, 6, 7, 9, 11, 13, 14, 16}	B	G	I	D	D
{4, 6, 7, 9, 11, 13, 14, 16}	C	G	I	D	D
{.}	D	D	D	D	D
{27, 31, 32, 34, 36, 38, 40, 42, 44}	E	D	D	J	K
{29, 31, 32, 34, 36, 38, 40, 42, 44}	F	D	D	J	K
{7, 8, 9, 11, 12, 13, 14, 16}	G	G	I	D	D
{9, 10, 11, 12, 13, 14, 15, 16, 18, 19, 21, 23, 25, 44}	H	L	M	D	D
{17, 18, 19, 21, 23, 25, 44}	I	D	O	D	D
{32, 33, 34, 36, 37, 38, 40, 42, 44}	J	D	N	J	K
{32, 34, 35, 36, 37, 38, 40, 42, 44}	K	D	M	J	K
{9, 10, 11, 12, 13, 14, 15, 16, 18, 19, 21, 23, 25, 44}	L	G	L	D	D
{12, 18, 19, 21, 22, 23, 24, 25, 44}	M	D	O	D	D
{19, 20, 21, 23, 25, 44}	N	N	O	D	D
{19, 21, 22, 23, 25, 44}	O	D	O	D	D



h) [ae03] + @[ae03]+.(com|net|org)(. (gt|cr|co))?

